Geometric certificates for the Frohmanian tether: Lie–Poisson structure, the metriplectic tethered sum, C2 degeneracy, and open items under C1–C3

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Abstract

We list the Lean 4/Mathlib certificates in the repository `frohmanian-symplectic-tether-may-form` that bear on the geometry of the tethered bracket. Proved: the field-level classical bracket on finite Fourier sums is antisymmetric and satisfies the Jacobi identity, hence is a Poisson bracket; that its symplectic leaves are the coadjoint orbits is standard background and is not formalized. The tethered sum $\{\cdot,\cdot\}_{\mathrm{LP}} + T$ splits uniquely into the antisymmetric classical part and the symmetric tether T, which is degenerate in the energy direction and negative semidefinite: it is metriplectic, and its Jacobi identity fails on explicit divergence-free test fields. Condition C2 (degeneracy on the kinetic-energy gradient u) forces every symmetric correction to see only Π_u-components, separating the conservative skeleton (span of u, where the bracket is the classical one) from the dissipative directions $u^{\perp}$. Open under C1–C3 (matching the repository README): C3 enters the Lean only as an explicit hypothesis; the quartic weight is not forced by C1–C3; and no damping estimate is proved or claimed. No claim is made about Navier–Stokes regularity or the Clay Millennium Problem. All listed theorems build with no `sorry` and depend only on `propext`, `Classical.choice`, `Quot.sound`.

Setting

$T(F,G) = -\kappa w \langle \Pi_u F, \Pi_u G\rangle$, $\Pi_u x = x - \frac{\langle u,x\rangle}{\|u\|^2} u$ (`Basic.lean`: `frohmanianTether`, `piU`). Abstract model: real inner-product space H. Field model (`FieldJacobi.lean`): finite Fourier sums `Fld` on $\mathbb{T}^3$ with bracket `br`, pairing `pr`, tether `tetF`.

Certificates

G1. Classical part is Poisson. `FieldJacobi.br_antisymm` ($\mathrm{br}\, a\, b = -\mathrm{br}\, b\, a$), `FieldJacobi.jacF_eq_zero` (Jacobi for every a, b, c). Abstract: `jacobi_classical_cancel`. *Background, not formalized:* symplectic leaves of the Lie–Poisson bracket are coadjoint orbits (Kirillov–Kostant–Souriau); C1 is modelled only by isometries fixing u (`InvariantUnderCoadjointAction`).

G2. Tethered sum is metriplectic. `Metriplectic.lean`: `split_unique`, `antiPart_frohmanianTether` (T has no antisymmetric part), `metripllecticBracket_split` (antisymmetric part $= P$, symmetric part $= T$), `metriplectic_energy_conserved`. `Basic.frohmanianTether_self_nonpos`. Field level (`FieldMetriplectic.lean`): `tetF_symm`, `tetF_energy_left/right`, `dissF_nonpos` (real fields, $\kappa w \geq 0$). Jacobi fails: `jacobi_iff_mixed_cancel`; `FieldJacobi.shearX_jacobi_fails`,

`shearY_jacobi_fails_transverse` ($\kappa w \neq 0$); `shearY_jacobi_holds` on the shear class. Self-bracket $\{F, F\} = T(F, F) \neq 0$: `TetheredSelfBracket.lean` (Zenodo draft 10.5281/zenodo.23965512).

G3. C2 separates skeleton from dissipative directions. `projectionOrthogonalToU` ($\Pi_u u = 0$), `frohmanianTether_degenerate`, `degeneracy_forces_projection` (C2 $\Rightarrow$ $B(F, G) = B(\Pi_u F, \Pi_u G)$ for symmetric B), `tetheredBracket_kineticEnergy` ($\{F, H\}_{TF} = \{F, H\}_{\text{classical}}$), `admissible_eq_smul_frohmanianTether` ($\dim \geq 3$, symmetric + C1 + C2 $\Rightarrow$ $B = T[c, 1, u]$).

G4. Open under C1–C3. C3 is the parameter of `IsAdmissibleCorrection`; Route A (`RouteA.lean`, `frohmanian_symplectic_tether_theorem`) fixes $\kappa = S$ by matching, with S supplied. No theorem derives a damping term for the unmodified equation from C1–C3, and none refutes it; the quartic weight is not forced by C1–C3 (same open items as the repository README).

Not claimed: any damping estimate; Navier–Stokes regularity.